I had a reason for giving the name *c* to our constant—the symbol *c* often stands for the speed of light. So is this particle a photon? No, not really. But our description of this hypothetical particle is pretty close to the correct description of a neutrino that moves at the speed of light. (Real neutrinos probably move at a speed that is immeasurably smaller than the speed of light.) This Hamiltonian would be a very good description of a one-dimensional neutrino except for one problem: the particle described by our wave function can only move to the right. To round out this description, we would have to add another possibility—that the particle could also move to the left![2]

Our right-going zaxon[3] has another oddball feature—its energy can be either positive or negative. This is because the **P** operator, as a vector, can take on positive or negative values. In general, the energy of a particle with negative momentum is negative, and the energy of a particle with positive momentum is positive. I won't say more about this except that the problem of negative energy for this kind of particle was solved by Dirac, who used it to establish the theoretical basis for antiparticles. For our purposes, we can ignore this issue and simply allow the energy of our particle to be either positive or negative.

Since the wave function of our particle moves rigidly down the *x* axis, so does the probability distribution. As a result, the expectation value of *x* moves in exactly the same way, which is to say that it moves to the right with velocity *c*. That's the essential quantum mechanics of this system. However, there is another important thing to keep in mind. When we said the velocity *c* is a *fixed* constant, we weren't kidding. Our particle can only exist in a state where it moves at this particular velocity. It can never slow down or speed up.

How does this compare with the classical description of such a particle? Starting with the same Hamiltonian, a classical physicist would just write Hamilton's equations. With $H = cP$, Hamilton's equations are